Open-Set Recognition of Plant Species: Detecting Unfamiliar Flora for Biodiversity Screening

Group 016

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Team Member Contributions

- **Aswath Oruganti:** Implemented and trained the EfficientNet-B0 classification pipeline, developed the baseline image classification experiments, and contributed to model evaluation.
- **Bakiye Hilal Khaniyev:** Designed the open-set evaluation protocol, implemented the held-out species testing pipeline and embedding extraction, performed representation analysis, and wrote the final report.
- **Yussif Salama:** Implemented and evaluated the open-set recognition methods (Mahalanobis Distance, One-Class SVM, Isolation Forest), performed ROC/AUC analysis, and compared the open-set detection approaches.

1 Introduction

Accurate identification of plant species from photographs has become increasingly important for biodiversity monitoring, ecological research, conservation efforts, and citizen science applications. Recent advances in deep learning, particularly convolutional neural networks (CNNs), have substantially improved automated plant classification, with several publicly available models now achieving high accuracy on large-scale image datasets. These systems are increasingly being integrated into mobile applications and biodiversity platforms to assist both researchers and the general public in species identification.

Despite these advances, most existing image classification systems operate under the closed-set assumption: every test image is assumed to belong to one of the species observed during training. In practice, however, this assumption rarely holds. Natural ecosystems contain many species that are absent from the training data, whether because they are rare, newly introduced, geographically localized, or simply underrepresented in existing datasets. Conventional classifiers are nevertheless forced to assign these unfamiliar observations to one of the known classes, often producing highly confident but incorrect predictions.

This limitation presents a significant challenge for biodiversity screening. In many ecological applications, recognizing that a specimen is unfamiliar can be more valuable than assigning an incorrect species label. Flagging observations that fall outside the model's learned knowledge allows experts to prioritize potentially novel or unusual specimens for further investigation instead of spending effort correcting erroneous predictions.

This project investigates the problem of open-set recognition for plant species identification. Rather than attempting to classify every possible plant species, we train image classification models on a curated subset of well-represented species and evaluate their ability to distinguish between known and previously unseen species. The study combines deep feature extraction using modern convolutional neural networks with several open-set recognition techniques operating in the learned embedding space.

Specifically, we compare feature representations learned by ResNet-18 and EfficientNet-B0 and evaluate multiple strategies for detecting unfamiliar species, including Maximum Softmax Probability, Mahalanobis distance, One-Class Support Vector Machines, and Isolation Forest. The objective is not only to obtain high classification accuracy on known species but also to determine which open-set recognition methods most effectively identify species that were intentionally excluded from the training process. Such a system could provide a practical first-pass screening tool for biodiversity monitoring by automatically identifying observations that warrant expert review.

2 Related Work

Deep convolutional neural networks have become the dominant approach for image-based plant species identification, largely due to the availability of large annotated datasets and advances in transfer learning. The Pl@ntNet-300K dataset introduced by Garcin et al. [1] is one of the largest publicly available plant image datasets and has become a standard benchmark for evaluating plant classification algorithms. However, the dataset exhibits a highly long-tailed class distribution, making accurate recognition of infrequently observed species particularly challenging.

Transfer learning has significantly improved image classification performance by allowing models pretrained on ImageNet to learn domain-specific representations using relatively limited task-specific data. Residual Networks (ResNet) [2] introduced skip connections that enable the successful training of deep neural networks, while EfficientNet [7] demonstrated that carefully balancing network depth, width, and input resolution can substantially improve both predictive accuracy and computational efficiency. Both architectures have been successfully applied across numerous fine-grained image classification tasks, including biodiversity monitoring.

While these models achieve strong performance under the closed-set assumption, they remain unable to recognize observations belonging to previously unseen classes. Open-set recognition addresses this limitation by enabling models to identify inputs that fall outside the training distribution. Hendrycks and Gimpel [3] proposed Maximum Softmax Probability (MSP) as a simple baseline for out-of-distribution detection using classification confidence. More advanced approaches exploit the geometry of the learned feature space, such as Mahalanobis distance [4], or treat the problem as anomaly detection using methods including One-Class Support Vector Machines [6] and Isolation Forests [5]. These approaches have demonstrated improved robustness across a variety of open-set recognition tasks.

Building on this prior work, our study combines deep feature representations learned through transfer learning with multiple open-set recognition techniques to evaluate their effectiveness for biodiversity screening. Rather than focusing solely on closed-set classification accuracy, we emphasize the ability to reliably identify plant species that were intentionally excluded from the training process.

3 Dataset and Data Preparation

3.1 PlantNet-300K Dataset

This study used the Pl@ntNet-300K dataset [1], a large-scale public benchmark for plant image classification containing 306,146 images spanning 1,081 plant species. The images were collected through the Pl@ntNet citizen science platform and exhibit substantial variability in viewpoint, lighting conditions, background clutter, image quality, and developmental stage of the plants. Consequently, the dataset closely reflects the challenges encountered in real-world biodiversity monitoring.

One characteristic of Pl@ntNet-300K is its highly long-tailed class distribution. While a relatively small number of species are represented by thousands of images, many species contain only a handful of observations. Such imbalance makes supervised training difficult and can bias classification models toward the most frequently occurring species.

3.2 Dataset Curation

To obtain a sufficiently large and well-supported training set, we curated the dataset by selecting only the most image-rich species. Approximately the top 20% of species ranked by image count were retained, while species with very few observations were excluded from the supervised training process. This curation reduced the impact of severe class imbalance and ensured that each retained class contained enough samples for effective transfer learning.

Following curation, the retained species were randomly partitioned at the *species level* into two disjoint groups: *known* species and *unknown* species. Images belonging to the unknown species were completely removed from model training and validation. Consequently, the model never observed these species during learning, allowing them to serve as genuine unseen classes during evaluation.

For the known species, the dataset's predefined training, validation, and test splits were preserved. The training set was used for fine-tuning the convolutional neural networks, the validation set was used for model selection and hyperparameter tuning, and the held-out test set was reserved exclusively for the final evaluation.

The open-set evaluation dataset consisted of two components:

1. test images belonging to the known species, which should be correctly classified by the model, and

2. images belonging to the held-out unknown species, which should be identified as unfamiliar rather than assigned to a known class.

Constructing the evaluation dataset in this manner provides exact ground truth for both closed-set classification and open-set detection, while preventing information leakage between the training and testing stages.

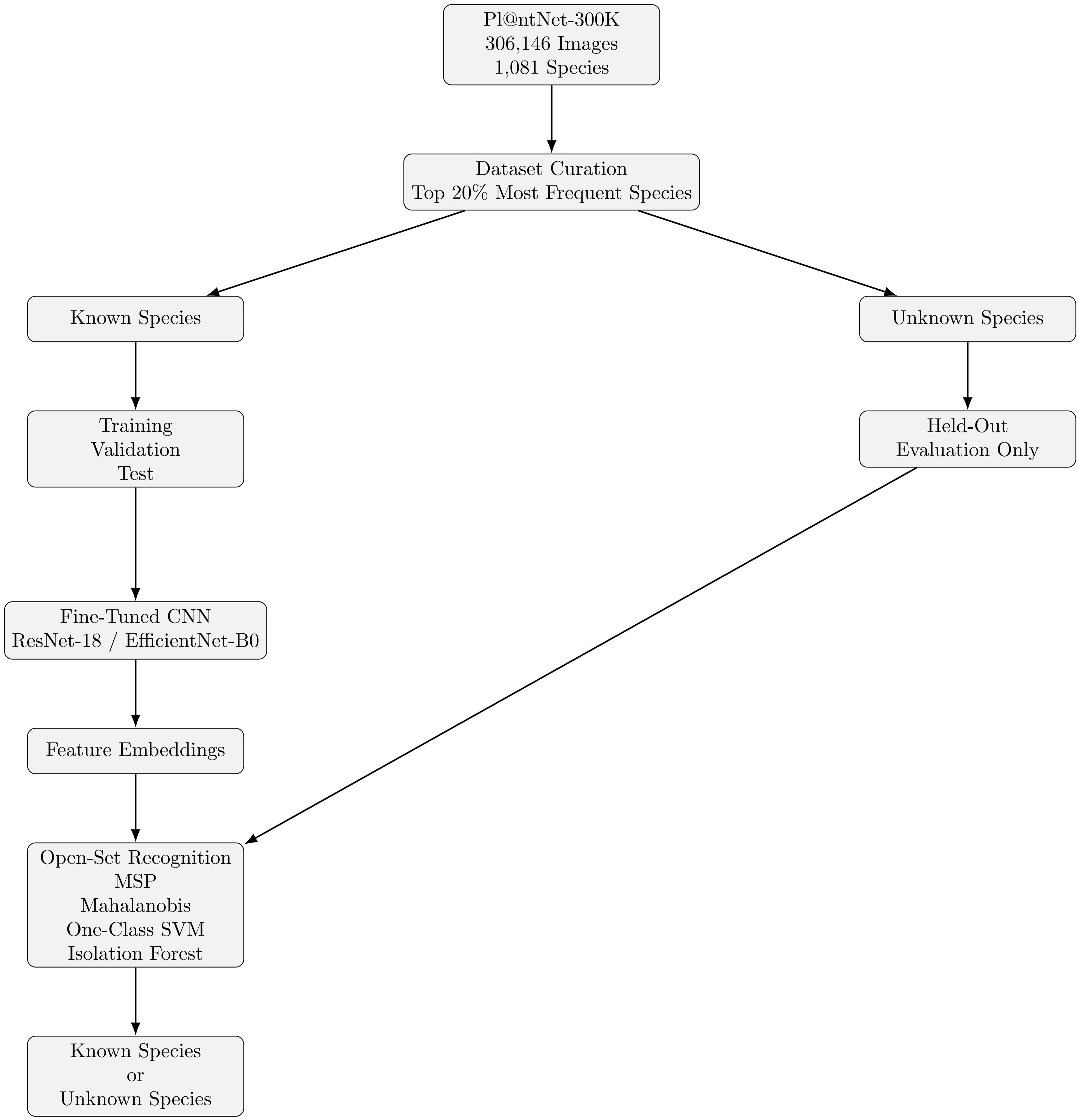

Figure 1: Overview of the proposed open-set recognition framework. The PlantNet-300K dataset is first curated to retain the most well-represented species. The retained species are divided into known and unknown groups. Only known species are used for model training, while unknown species are reserved exclusively for evaluation. Feature embeddings extracted from the fine-tuned CNN are used by several open-set recognition methods to determine whether an input belongs to a known species or should be flagged as unfamiliar.

4 Methodology

Our proposed framework combines supervised image classification with open-set recognition to identify plant species that were not observed during training. The overall workflow is illustrated in Figure 1. A convolutional neural network is first fine-tuned to classify the known plant species while simultaneously learning discriminative feature representations. These learned embeddings are then used by several open-set recognition algorithms to determine whether a test image belongs to one of the known species or should instead be classified as unfamiliar.

4.1 Feature Learning

Two convolutional neural network architectures were investigated in this study: ResNet-18 [2] and EfficientNet-B0 [7]. Both models were initialized with ImageNet-pretrained weights and fine-tuned on the curated set of known plant species using transfer learning.

Beyond their role as classifiers, both networks served as feature extractors. After training, feature vectors were obtained from the penultimate layer for every image. These embeddings provide a compact representation of visual characteristics and form the basis for all subsequent open-set recognition methods.

4.2 Open-Set Evaluation Protocol

Species assigned to the unknown set were completely excluded from training and validation. During evaluation, the test dataset consisted of two groups:

1. Images belonging to known species, which should be correctly classified.
2. Images belonging to unknown species, which should be identified as unfamiliar.

This evaluation protocol ensures that unknown species remain entirely unseen during model training, thereby preventing information leakage and providing an unbiased assessment of open-set recognition performance.

4.3 Open-Set Recognition Methods

To distinguish familiar and unfamiliar species, several complementary open-set recognition methods were investigated.

4.3.1 Maximum Softmax Probability

Maximum Softmax Probability (MSP) [3] was used as a simple baseline. The highest softmax probability produced by the classifier was interpreted as its confidence. Images with confidence below a predefined threshold were classified as unfamiliar.

4.3.2 Mahalanobis Distance

Mahalanobis Distance models the distribution of each known class within the learned embedding space. The distance between a test embedding and the nearest class distribution is computed, and samples whose distance exceeds a threshold are classified as unknown.

4.3.3 One-Class Support Vector Machine

One-Class Support Vector Machine (OC-SVM) [6] is trained exclusively on embeddings extracted from known species. The algorithm estimates the support of the known data distribution and flags observations falling outside this region as unfamiliar.

4.3.4 Isolation Forest

Isolation Forest [5] approaches the problem as anomaly detection by recursively partitioning the embedding space. Samples that are isolated using relatively few partitions receive higher anomaly scores and are classified as unknown.

5 Experimental Setup

All experiments were implemented in Python using the PyTorch deep learning framework. Two convolutional neural network architectures, ResNet-18 and EfficientNet-B0, were initialized with ImageNet-pretrained weights and fine-tuned on the curated set of known plant species using transfer learning.

Images were resized and normalized according to the preprocessing requirements of the pre-trained models. The dataset was divided into training, validation, and testing sets using the predefined splits provided by Pl@ntNet-300K for the known species, while all images belonging to the unknown species were reserved exclusively for open-set evaluation.

Model selection was performed using the validation set. After training, the best-performing model checkpoint was retained for evaluation on the held-out test set. In addition to classification accuracy, feature embeddings from the penultimate layer of each network were extracted for subsequent representation analysis and open-set recognition experiments.

To ensure reproducibility, a fixed random seed was used throughout all experiments. During training, the models were optimized using stochastic gradient descent with transfer learning from ImageNet-pretrained weights. Validation accuracy was monitored after each epoch, and the checkpoint achieving the highest validation accuracy was selected for subsequent evaluation. Once training was complete, feature embeddings extracted from the penultimate layer were used for both representation analysis and the open-set recognition methods.

6 Results

6.1 Closed-Set Classification Performance

We first evaluated the classification performance of the two convolutional neural network architectures on the known species. ResNet-18 served as a lightweight baseline model, while EfficientNet-B0 was investigated as a more computationally sophisticated architecture capable of learning richer feature representations.

Both models converged successfully during fine-tuning and achieved strong classification performance on the validation data. EfficientNet-B0 obtained the highest validation accuracy, whereas ResNet-18 required substantially less training time and computational resources. This trade-off between predictive performance and computational efficiency motivated the use of both models in the subsequent representation analysis.

Table 1 summarizes the performance of the two architectures.

Table 1: Classification performance of the fine-tuned convolutional neural networks.

Model	Validation Accuracy	Training Time
ResNet-18	72.3%	∼385 min
EfficientNet-B0	78.2%	∼6780 min

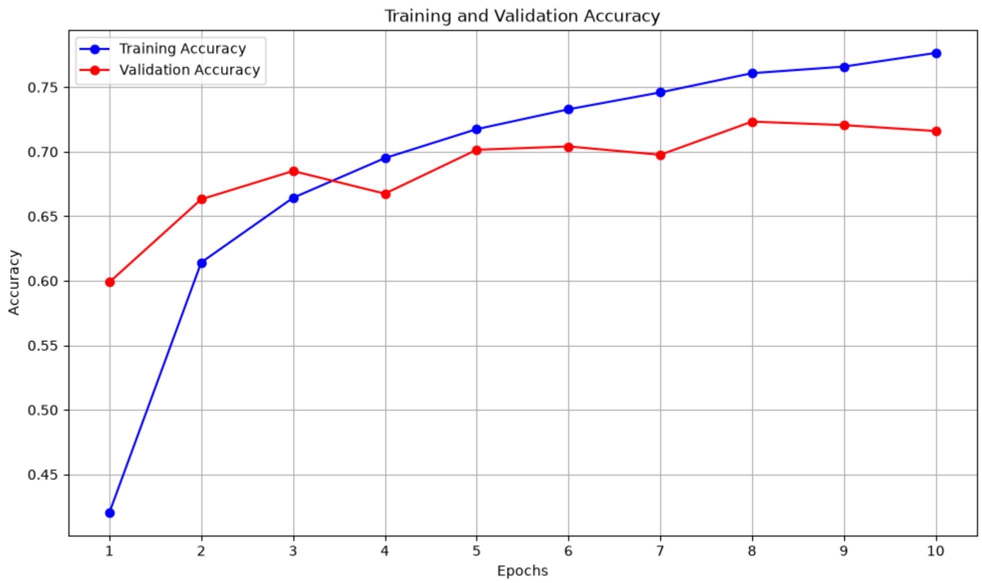

Figure 2: Training and validation accuracy of ResNet-18 during fine-tuning.

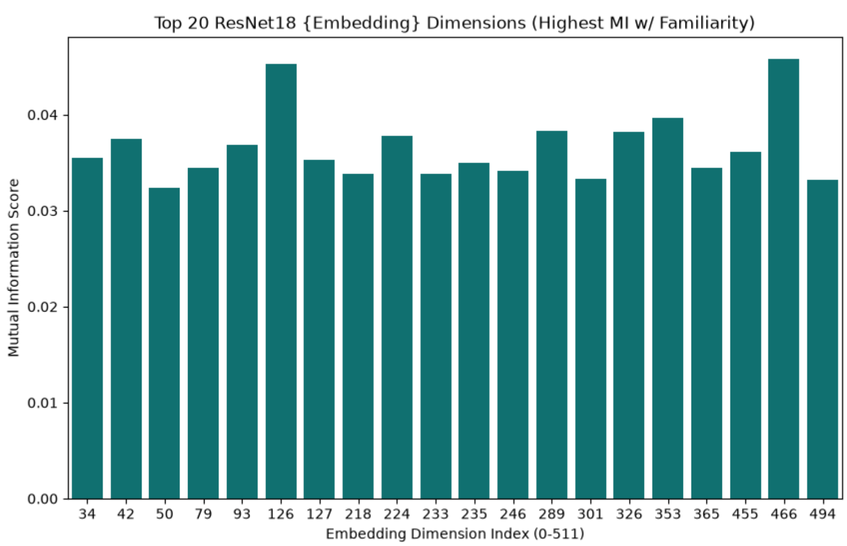

Figure 3: Mutual Information Scores - After Fine-Tuning ResNet18

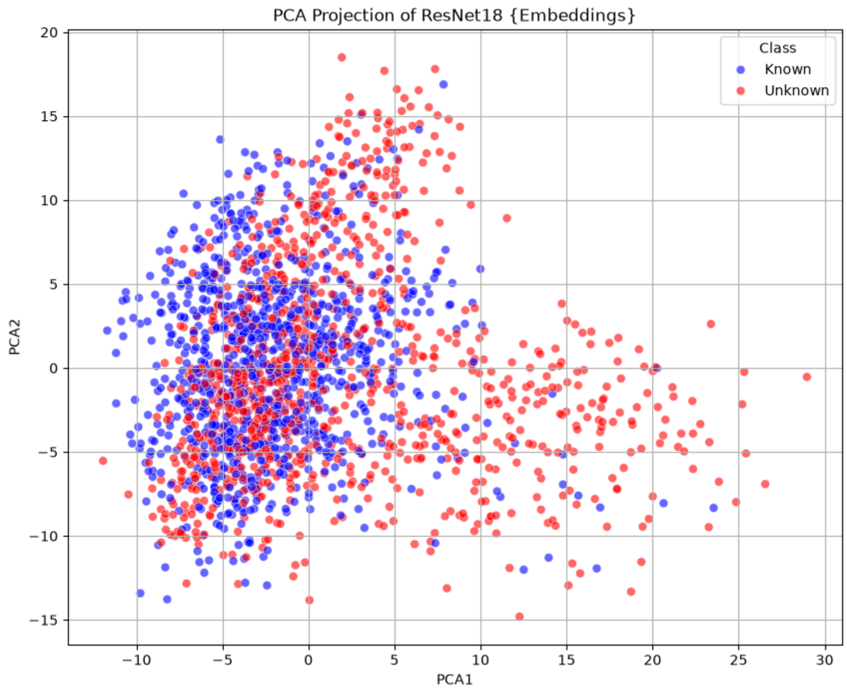

Figure 4: PCA projection - After Fine-Tuning ResNet18

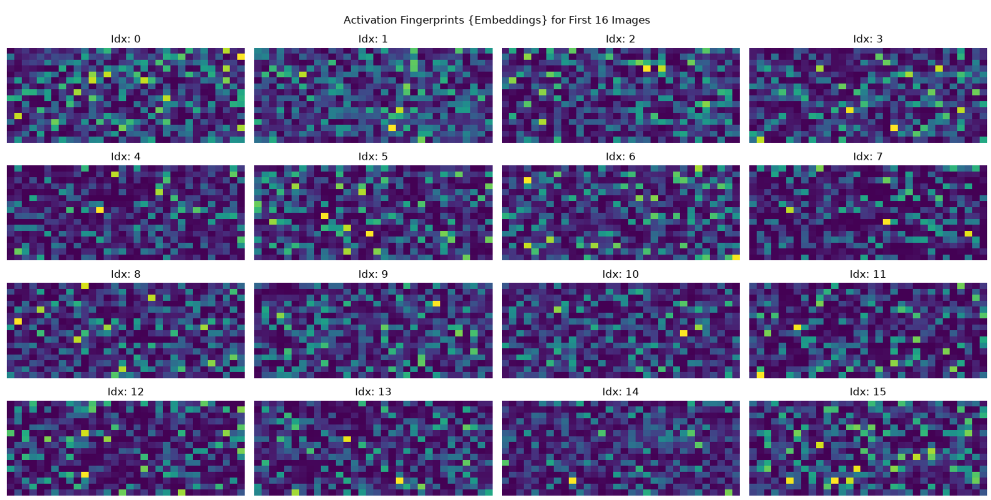

Figure 5: Activation Fingerprints - After Fine-Tuning ResNet18

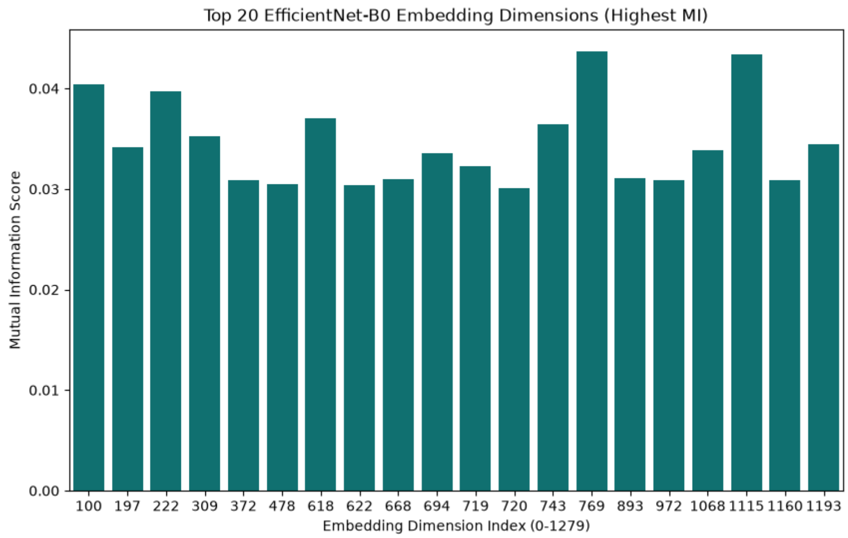

Figure 6: Mutual Information Scores - After Fine-Tuning EfficientNetB0

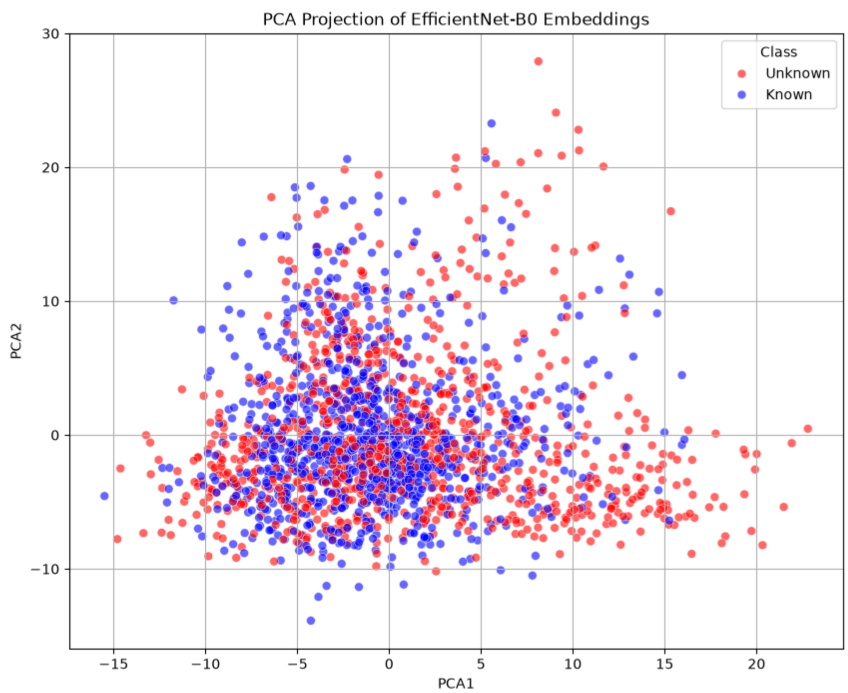

Figure 7: PCA projection - After Fine-Tuning EfficientNetB0

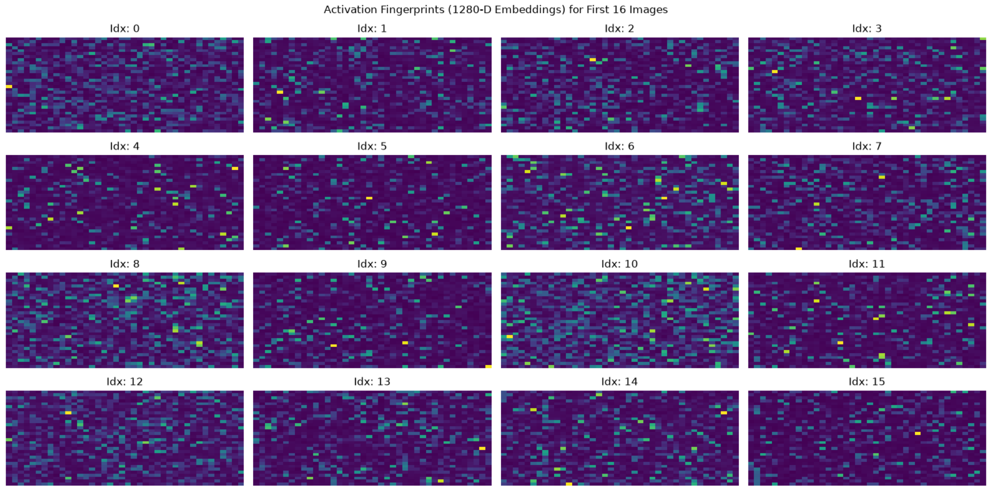

Figure 8: Activation Fingerprints - After Fine-Tuning EfficientNetB0

6.2 Representation Analysis

Although the overall classification accuracy provides a useful measure of predictive performance, it does not fully characterize the quality of the learned feature representations. To better understand how each network organizes plant images in the embedding space, we analyzed the extracted feature vectors using several complementary visualization techniques.

First, we measured the mutual information between the individual embedding dimensions and the class labels. This analysis identifies the features that contribute the most strongly to species discrimination. We additionally projected the high-dimensional embeddings into two dimensions using Principal Component Analysis (PCA) to visualize the overall class structure learned by each network. Finally, activation fingerprint visualizations were generated to compare the internal representations learned by the two architectures.

Together, these analyzes provide qualitative information on the separability and discriminative power of the learned feature spaces, complementing the quantitative classification metrics.

6.3 Open-Set Recognition Performance

6.4 Discussion

7 Limitations

Several limitations should be considered when interpreting the results of this study. First, the evaluation was conducted using a curated subset of the Pl@ntNet-300K dataset, meaning that the learned models were exposed only to relatively well-represented species. Although this improves training stability, it does not fully reflect the long-tailed distribution encountered in real-world biodiversity monitoring.

Second, all unknown species originated from the same dataset as the known species. While this provides a controlled evaluation protocol, future work should investigate generalization to completely independent datasets collected under different environmental conditions.

Finally, the open-set recognition methods operate exclusively on visual information. Incorporating additional metadata, such as geographic location, seasonality, or taxonomic relationships, could further improve the detection of unfamiliar species.

8 Conclusion

This project investigated open-set recognition for plant species identification using deep feature representations learned from convolutional neural networks. A curated subset of the Pl@ntNet-300K dataset was used to train image classification models while intentionally excluding a subset of species to simulate unfamiliar observations during evaluation.

Transfer learning with ResNet-18 and EfficientNet-B0 produced discriminative feature embeddings suitable for both classification and open-set recognition. Beyond evaluating classification performance, representation analyses were conducted to better understand the structure of the learned embedding spaces. These analyses provided insight into the separability of plant species and established a foundation for evaluating multiple open-set recognition techniques.

Overall, this work demonstrates that combining deep feature extraction with open-set recognition offers a practical framework for biodiversity screening. Rather than forcing every observation into a known category, the proposed system can identify unfamiliar species for expert review, making automated plant identification more reliable in real-world ecological applications. Future work will investigate additional feature representations and more advanced open-set recognition methods to further improve robustness when encountering previously unseen species.

References

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